

HW2

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Note: We use $\subset \equiv \subseteq$ and $\supset \equiv \supseteq$ for the purposes of set relations. This note will be dropped henceforth in future problemsets.

Q1: A.29

Let $\mathbb{S} = C[0, 1]$ and $d_p(f, g) \equiv \left(\int_0^1 |f(t) - g(t)|^p dt \right)^{\frac{1}{p}}$ for $1 \leq p < \infty$ and $d_\infty(f, g) = \sup\{|f(t) - g(t)| : t \in [0, 1]\}$.

Note: May use the inequality:

$$|a + b|^p \leq 2^p (|a|^p + |b|^p)$$

(a)

Let $f(x) \equiv 1$. Let $f_n(t) \equiv 1$ for $0 \leq t \leq 1 - \frac{1}{n}$, and $f_n(t) = n(1 - t)$ for $1 - \frac{1}{n} \leq t \leq 1$. Show that $d_p(f_n, f) \rightarrow 0$ for $1 \leq p < \infty$ but $d_\infty(f_n, f) \not\rightarrow 0$.

Answer

Sketch: The difference vanishes on $[0, 1 - 1/n]$, so $d_p(f_n, f)^p$ reduces to an integral over the ramp $[1 - 1/n, 1]$, which is $O(1/n) \rightarrow 0$. For d_∞ , the difference at $t = 1$ is 1 for every n , which rules out convergence.

Proof.

Case $1 \leq p < \infty$. On $[0, 1 - 1/n]$, $f_n(t) = 1 = f(t)$, so $|f_n(t) - f(t)|^p = 0$ there. On $[1 - 1/n, 1]$, $f_n(t) - f(t) = n(1 - t) - 1$. Hence

$$d_p(f_n, f)^p = \int_0^1 |f_n(t) - f(t)|^p dt = \int_{1-\frac{1}{n}}^1 |n(1 - t) - 1|^p dt.$$

Substitute $u = 1 - t$ ($du = -dt$; $t = 1 - 1/n \Rightarrow u = 1/n$, $t = 1 \Rightarrow u = 0$):

$$= \int_0^{1/n} |nu - 1|^p du.$$

Substitute $w = nu - 1$ ($dw = n du$; $u = 0 \Rightarrow w = -1$, $u = 1/n \Rightarrow w = 0$):

$$= \frac{1}{n} \int_{-1}^0 |w|^p dw = \frac{1}{n} \int_0^1 w^p dw = \frac{1}{n(p+1)}.$$

Since p is fixed, $\frac{1}{n(p+1)} \rightarrow 0$ as $n \rightarrow \infty$, so $d_p(f_n, f)^p \rightarrow 0$, hence $d_p(f_n, f) \rightarrow 0$, for every $1 \leq p < \infty$.

Case $p = \infty$. On $[1 - 1/n, 1]$, $n(1 - t)$ is a strictly decreasing affine function of t , taking the value $n(1 - (1 - 1/n)) = 1$ at $t = 1 - 1/n$ and the value 0 at $t = 1$. So $|n(1 - t) - 1|$ increases monotonically from 0 to 1 on this interval, attaining its maximum of 1 at $t = 1$:

$$|f_n(1) - f(1)| = |n(1 - 1) - 1| = |0 - 1| = 1.$$

Since the difference is 0 on $[0, 1 - 1/n]$ and its supremum on $[1 - 1/n, 1]$ is 1 (attained at $t = 1$),

$$d_\infty(f_n, f) = \sup_{t \in [0, 1]} |f_n(t) - f(t)| = 1$$

for every $n \geq 1$. Hence $\lim_{n \rightarrow \infty} d_\infty(f_n, f) = 1 \neq 0$, so $d_\infty(f_n, f) \not\rightarrow 0$. //

(b)

Fix $f \in C[0, 1]$. Let $g_n(t) = f(t)$, $0 \leq t \leq 1 - \frac{1}{n}$, and $g_n(t) = f(1 - \frac{1}{n}) + (f(1) - f(1 - \frac{1}{n}))n(t + \frac{1}{n} - 1)$, $1 - \frac{1}{n} \leq t \leq 1$.

Show that $d_p(g_n, f) \rightarrow 0$ for all $1 \leq p \leq \infty$.

Answer

Sketch: $g_n(t) - f(t)$ is a convex combination of $f(1 - 1/n) - f(t)$ and $f(1) - f(t)$ on the ramp $[1 - 1/n, 1]$, and 0 elsewhere. For finite p , the Note plus boundedness of f gives $d_p(g_n, f)^p = O(1/n) \rightarrow 0$. For $p = \infty$, uniform continuity of f does the job directly, since the ramp shrinks to the point $t = 1$.

Proof.

Fix $f \in C[0, 1]$. Since $[0, 1]$ is compact and f is continuous, f is bounded: $\exists M > 0$ such that $|f(t)| \leq M$ for all $t \in [0, 1]$ (Extreme Value Theorem).

For $t \in [1 - 1/n, 1]$, let $s \equiv n(t + 1/n - 1) \in [0, 1]$, the linear reparametrization of $[1 - 1/n, 1]$ onto $[0, 1]$. By definition $g_n(t) = (1 - s)f(1 - 1/n) + sf(1)$, so

$$g_n(t) - f(t) = (1 - s)(f(1 - \frac{1}{n}) - f(t)) + s(f(1) - f(t)). \quad (*)$$

(Check: $(1 - s)f(1 - \frac{1}{n}) + sf(1) - (1 - s)f(t) - sf(t) = g_n(t) - f(t)$, using $(1 - s) + s = 1$.) On $[0, 1 - 1/n]$, $g_n(t) = f(t)$, so the difference is 0 there.

Case $1 \leq p < \infty$. By (*) and the Note, $|a + b|^p \leq 2^p(|a|^p + |b|^p)$, with $a = (1 - s)(f(1 - \frac{1}{n}) - f(t))$ and $b = s(f(1) - f(t))$:

$$|g_n(t) - f(t)|^p \leq 2^p \left[(1 - s)^p |f(1 - \frac{1}{n}) - f(t)|^p + s^p |f(1) - f(t)|^p \right].$$

Since $s \in [0, 1]$, $(1 - s)^p \leq 1$ and $s^p \leq 1$, so

$$|g_n(t) - f(t)|^p \leq 2^p \left[|f(1 - \frac{1}{n}) - f(t)|^p + |f(1) - f(t)|^p \right].$$

Since $|f(1 - \frac{1}{n})|, |f(t)|, |f(1)| \leq M$, both $|f(1 - \frac{1}{n}) - f(t)| \leq 2M$ and $|f(1) - f(t)| \leq 2M$, giving

$$|g_n(t) - f(t)|^p \leq 2^p [(2M)^p + (2M)^p] = 2^{p+1} (2M)^p$$

for all $t \in [1 - 1/n, 1]$. Hence

$$d_p(g_n, f)^p = \int_0^1 |g_n(t) - f(t)|^p dt = \int_{1 - \frac{1}{n}}^1 |g_n(t) - f(t)|^p dt \leq 2^{p+1} (2M)^p \cdot \frac{1}{n}.$$

Since p is fixed, $2^{p+1}(2M)^p$ is a constant, so $d_p(g_n, f)^p \rightarrow 0$ as $n \rightarrow \infty$, hence $d_p(g_n, f) \rightarrow 0$.

Case $p = \infty$. Since f is continuous on the compact set $[0, 1]$, f is uniformly continuous (Heine–Cantor Theorem): $\forall \varepsilon > 0, \exists \delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$ for all $x, y \in [0, 1]$.

Let $\varepsilon > 0$ be given, and choose $N \in \mathbb{N}$ with $1/N < \delta$. Let $n \geq N$ and $t \in [1 - 1/n, 1]$ be given. Since $|t - (1 - \frac{1}{n})| \leq \frac{1}{n} \leq \frac{1}{N} < \delta$ and $|t - 1| \leq \frac{1}{n} \leq \frac{1}{N} < \delta$,

$$|f(1 - \frac{1}{n}) - f(t)| < \varepsilon \quad \text{and} \quad |f(1) - f(t)| < \varepsilon.$$

By (*) and the triangle inequality (the Note only covers finite p),

$$|g_n(t) - f(t)| \leq (1 - s)|f(1 - \frac{1}{n}) - f(t)| + s|f(1) - f(t)| < (1 - s)\varepsilon + s\varepsilon = \varepsilon$$

for all $t \in [1 - 1/n, 1]$. Combined with $g_n(t) - f(t) = 0$ on $[0, 1 - 1/n]$,

$$d_\infty(g_n, f) = \sup_{t \in [0, 1]} |g_n(t) - f(t)| \leq \varepsilon$$

for all $n \geq N$. Since $\varepsilon > 0$ was arbitrary, $d_\infty(g_n, f) \rightarrow 0$.

Combining both cases, $d_p(g_n, f) \rightarrow 0$ for all $1 \leq p \leq \infty$. //

Q2: A.34

Show that unions of open sets are open and intersection of any two open sets is open. Give an example to show that the intersection of an infinite number of open sets need not be open.

Answer

Sketch: Parts 1–2 argue directly from the ε -ball definition: pull a radius from each set containing a point, then take the max (resp. min). Part 3 exhibits $\bigcap_{n \geq 1} (-1/n, 1/n) = \{0\}$, which fails the open-set definition outright.

Proof.

Part 1: Unions of open sets are open.

Let $(\mathbb{S}, d)$ be a metric space, I an index set, and $\{O_\alpha\}_{\alpha \in I}$ a family of open sets on $\mathbb{S}$. Let $x \in \bigcup_{\alpha \in I} O_\alpha$ be given. Then $\exists \alpha_0 \in I$ such that $x \in O_{\alpha_0}$. Since O_{α_0} is open, $\exists \delta > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta \Rightarrow y \in O_{\alpha_0}$. Since $O_{\alpha_0} \subset \bigcup_{\alpha \in I} O_\alpha$, this gives $d(x, y) < \delta \Rightarrow y \in \bigcup_{\alpha \in I} O_\alpha$. Since $x \in \bigcup_{\alpha \in I} O_\alpha$ was arbitrary, $\bigcup_{\alpha \in I} O_\alpha$ is open.

Part 2: The intersection of any two open sets is open.

Let A, B be open sets on $(\mathbb{S}, d)$. Let $x \in A \cap B$ be given, i.e., $x \in A$ and $x \in B$. Since A is open, $\exists \delta_1 > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta_1 \Rightarrow y \in A$. Since B is open, $\exists \delta_2 > 0$ such that for all $y \in \mathbb{S}$, $d(x, y) < \delta_2 \Rightarrow y \in B$. Let $\delta \equiv \min(\delta_1, \delta_2) > 0$. Then for all $y \in \mathbb{S}$ with $d(x, y) < \delta$, both $d(x, y) < \delta_1$ and $d(x, y) < \delta_2$ hold, so $y \in A$ and $y \in B$, i.e., $y \in A \cap B$. Since $x \in A \cap B$ was arbitrary, $A \cap B$ is open.

Part 3: The intersection of infinitely many open sets need not be open.

Let $A_n \equiv (-1/n, 1/n)$ for each $n \geq 1$, an open subset of $(\mathbb{R}, |\cdot|)$. We claim $\bigcap_{n \geq 1} A_n = \{0\}$.

($\supset$) Since $0 \in (-1/n, 1/n)$ for every $n \geq 1$, $0 \in \bigcap_{n \geq 1} A_n$.

($\subset$) Let $x \in \bigcap_{n \geq 1} A_n$ be given, and suppose for contradiction $x \neq 0$. By the Archimedean property, $\exists N \in \mathbb{N}$ such that $1/N < |x|$, so $x \notin (-1/N, 1/N) = A_N$, contradicting $x \in \bigcap_{n \geq 1} A_n \subset A_N$. Hence $x = 0$.

Combining both containments, $\bigcap_{n \geq 1} A_n = \{0\}$.

Finally, $\{0\}$ is not open: for any $\delta > 0$, the point $y \equiv \delta/2$ satisfies $d(0, y) = \delta/2 < \delta$, yet $y \neq 0$, so $y \notin \{0\}$. Thus no δ -ball centered at 0 is contained in $\{0\}$, so $\{0\}$ is not open.

Hence $\bigcap_{n \geq 1} A_n = \{0\}$ is a countable (in fact infinite) intersection of open sets that is not open. //

Q3: A.36

Let $f_n(x) = x^n$ and $g(x) \equiv 0$ on $\mathbb{R}$. Then $\{f_n\}_{n \geq 1}$ converges pointwise to g on $(-1, 1)$, uniformly on $[a, b]$ for $-1 < a < b < 1$, but not uniformly on $(0, 1)$.

Answer

Sketch: Parts (a)–(b) use the same bound $|x|^n < \varepsilon \iff n > \log \varepsilon / \log |x|$; on $[a, b]$, bounding $|x|$ by $M \equiv \max(|a|, |b|) < 1$ makes N independent of x , giving uniformity. Part (c) negates the uniform convergence definition directly, using $x_N = (1/2)^{1/N}$ as a witness at each N .

Proof.

(a) Pointwise convergence on $(-1, 1)$.

Let $x \in (-1, 1)$ be given. If $x = 0$, then $f_n(x) = 0 = g(x)$ for all $n \geq 1$, so convergence is immediate. If $x \neq 0$, then $|x| \in (0, 1)$, so $\log |x| < 0$. Let $\varepsilon > 0$ be given, and set

$$N \equiv \left\lfloor \frac{\log \varepsilon}{\log |x|} \right\rfloor + 1.$$

For $n \geq N$, $n > \log \varepsilon / \log |x|$; since $\log |x| < 0$, multiplying both sides by $\log |x|$ reverses the inequality, giving $n \log |x| < \log \varepsilon$, i.e. $|x|^n < \varepsilon$. Thus $|f_n(x) - g(x)| = |x|^n < \varepsilon$ for all $n \geq N$.

In either case $\lim_{n \rightarrow \infty} f_n(x) = 0 = g(x)$. Since $x \in (-1, 1)$ was arbitrary, $\{f_n\}_{n \geq 1}$ converges pointwise to g on $(-1, 1)$.

(b) Uniform convergence on $[a, b]$ for $-1 < a < b < 1$.

Let $-1 < a < b < 1$ be given, and set $M \equiv \max(|a|, |b|)$. Since $a \neq b$, not both a, b equal 0, so $M \in (0, 1)$. For every $x \in [a, b]$, $|x| \leq M < 1$.

Let $\varepsilon > 0$ be given, and set

$$N_\varepsilon \equiv \left\lfloor \frac{\log \varepsilon}{\log M} \right\rfloor + 1.$$

By the same argument as in (a) (with M in place of $|x|$), $n \geq N_\varepsilon \implies M^n < \varepsilon$. Since $t \mapsto t^n$ is non-decreasing on $[0, \infty)$ and $0 \leq |x| \leq M$, $|x|^n \leq M^n$ for all $x \in [a, b]$. Thus, for all $x \in [a, b]$ and $n \geq N_\varepsilon$,

$$|f_n(x) - g(x)| = |x|^n \leq M^n < \varepsilon.$$

Since N_ε depends only on ε (via a, b) and not on x , $\{f_n\}_{n \geq 1}$ converges uniformly to g on $[a, b]$.

(c) Non-uniform convergence on $(0, 1)$.

We show the negation of uniform convergence holds: $\exists \varepsilon_0 > 0$ such that $\forall N \in \mathbb{N}$, $\exists n \geq N$ and $x \in (0, 1)$ with $|f_n(x) - g(x)| \geq \varepsilon_0$.

Let $\varepsilon_0 \equiv 1/2$. Let $N \in \mathbb{N}$ be given, and set $n \equiv N$ and $x_N \equiv (1/2)^{1/N}$; since $1/2 \in (0, 1)$, $x_N \in (0, 1)$ as well. Then

$$|f_n(x_N) - g(x_N)| = |x_N^N - 0| = \left(\left(\frac{1}{2} \right)^{1/N} \right)^N = \frac{1}{2} = \varepsilon_0 \geq \varepsilon_0.$$

Since $N \in \mathbb{N}$ was arbitrary, no N makes $|f_n(x) - g(x)| < \varepsilon_0$ hold for all $n \geq N$ and all $x \in (0, 1)$ simultaneously. Hence $\{f_n\}_{n \geq 1}$ does not converge uniformly to g on $(0, 1)$. //

Q4: 1.2

Let Ω be a finite set, i.e., the number of elements in Ω is finite. Let $\mathcal{F} \subset \mathcal{P}(\Omega)$ be an algebra. Show that $\mathcal{F}$ is a σ -algebra.

Answer

Sketch: By Proposition 1.1.1, it suffices to show $\mathcal{F}$ is closed under monotone unions. Since Ω is finite, an increasing sequence in $\mathcal{F}$ reduces the union to a finite union already in $\mathcal{F}$.

Proof. Since Ω is finite, write $m \equiv |\Omega| < \infty$; since $\mathcal{F} \subset \mathcal{P}(\Omega)$, every $A \in \mathcal{F}$ satisfies $A \subset \Omega$ and hence $|A| \leq m$.

Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ be given with $A_n \subset A_{n+1}$ for all $n \geq 1$. Since $A_n \subset A_{n+1}$, $|A_n| \leq |A_{n+1}|$ for all n , so $\{|A_n|\}_{n \geq 1}$ is a non-decreasing sequence in $\{0, 1, \dots, m\}$. A non-decreasing sequence of integers bounded above by m can increase at most m times, so there exists $N \in \mathbb{N}$ such that

$$|A_n| = |A_N| \quad \text{for all } n \geq N.$$

Fix $n \geq N$. By monotonicity, $A_N \subset A_n$; combined with $|A_N| = |A_n| < \infty$, this gives $A_N = A_n$ (a finite set contained in a finite set of equal cardinality must coincide with it). Thus $A_n = A_N$ for all $n \geq N$.

Consequently,

$$\bigcup_{n \geq 1} A_n = \bigcup_{n=1}^N A_n,$$

since for $n \leq N$, $A_n \subset A_N$ by monotonicity contributes nothing beyond A_N , and for $n > N$, $A_n = A_N$ contributes nothing new. As $\{A_1, \dots, A_N\} \subset \mathcal{F}$ is a finite collection and $\mathcal{F}$ is an algebra (Definition 1.1.1(c), closure under finite unions),

$\bigcup_{n=1}^N A_n \in \mathcal{F}$. Hence

$$\bigcup_{n \geq 1} A_n = A_N \in \mathcal{F}.$$

Since $\{A_n\}_{n \geq 1} \subset \mathcal{F}$ with $A_n \subset A_{n+1}$ was arbitrary, $\mathcal{F}$ is closed under monotone unions. As $\mathcal{F}$ is an algebra by hypothesis, Proposition 1.1.1 gives that $\mathcal{F}$ is a σ -algebra (Definition 1.1.2). //

Q5: 1.4

Let $\{\mathcal{F}_\theta : \theta \in \Theta\}$ be a family of σ -algebras on Ω . Show that $\mathcal{G} \equiv \bigcap_{\theta \in \Theta} \mathcal{F}_\theta$ is also a σ -algebra.

Answer

Sketch: Verify $\mathcal{G}$ satisfies Definition 1.1.2 directly: each defining property passes through the intersection unchanged, since it holds for every $\theta \in \Theta$.

Note: Definition 1.1.2 requires (a), (b), (c) — $\mathcal{F}$ an algebra — together with (d). But (a) and (b) alone give $\emptyset = \Omega^c \in \mathcal{F}$, so any finite union extends to a countable union by appending $\emptyset$'s; thus (d) already implies (c), and it suffices to verify (a), (b), (d).

Proof.

(a) Since each $\mathcal{F}_\theta$ is a σ -algebra, $\Omega \in \mathcal{F}_\theta$ for all $\theta \in \Theta$ (Definition 1.1.1(a)). Hence $\Omega \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

(b) Let $A \in \mathcal{G}$ be given. Then $A \in \mathcal{F}_\theta$ for every $\theta \in \Theta$. Since each $\mathcal{F}_\theta$ is a σ -algebra, $A^c \in \mathcal{F}_\theta$ for every $\theta \in \Theta$ (Definition 1.1.1(b)). Hence $A^c \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

(d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{G}$ be given. Then for every $\theta \in \Theta$, $A_n \in \mathcal{F}_\theta$ for all $n \geq 1$. Since each $\mathcal{F}_\theta$ is a σ -algebra, $\bigcup_{n \geq 1} A_n \in \mathcal{F}_\theta$ for every $\theta \in \Theta$ (Definition 1.1.2(d)). Hence $\bigcup_{n \geq 1} A_n \in \bigcap_{\theta \in \Theta} \mathcal{F}_\theta = \mathcal{G}$.

By (a), (b), and (d), and the Note above, $\mathcal{G}$ satisfies Definition 1.1.2. Hence $\mathcal{G}$ is a σ -algebra. //

Q6: 1.6

Let Ω be a nonempty set and let $\mathcal{A} \equiv \{A_i : i \in \mathbb{N}\}$ be a partition of Ω , i.e., $A_i \cap A_j = \emptyset$ for all $i \neq j$ and $\bigcup_{n \geq 1} A_n = \Omega$. Let $\mathcal{F} = \{\bigcup_{i \in J} A_i : J \subset \mathbb{N}\}$ where, for $J = \emptyset$, $\bigcup_{i \in J} A_i \equiv \emptyset$. Show that $\mathcal{F}$ is a σ -algebra.

Answer

Sketch: By the Note from Q5, it suffices to verify (a), (b), (d): (a) is immediate with $J = \mathbb{N}$; (b) follows by complementing the index set, using disjointness of the partition; (d) follows since countable unions correspond to unions of index sets.

Proof.

(a) Since $\bigcup_{n \geq 1} A_n = \Omega$, taking $J = \mathbb{N} \subset \mathbb{N}$ gives $\Omega = \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}$.

(b) Let $B \in \mathcal{F}$ be given. Then $B = \bigcup_{i \in J} A_i$ for some $J \subset \mathbb{N}$. Let $J^c \equiv \mathbb{N} \setminus J$. We claim $B^c = \bigcup_{i \in J^c} A_i$.

Since $\{A_i : i \in \mathbb{N}\}$ is a partition, $A_i \cap A_j = \emptyset$ for all $i \neq j$; in particular, for any $i \in J$ and $j \in J^c$, $i \neq j$, so $A_i \cap A_j = \emptyset$. Hence

$$\left(\bigcup_{i \in J} A_i \right) \cap \left(\bigcup_{j \in J^c} A_j \right) = \bigcup_{i \in J, j \in J^c} (A_i \cap A_j) = \emptyset.$$

Also, since $J \cup J^c = \mathbb{N}$,

$$\left(\bigcup_{i \in J} A_i \right) \cup \left(\bigcup_{j \in J^c} A_j \right) = \bigcup_{i \in \mathbb{N}} A_i = \Omega.$$

Thus $\bigcup_{i \in J} A_i$ and $\bigcup_{j \in J^c} A_j$ partition Ω , so

$$B^c = \Omega \setminus B = \Omega \setminus \bigcup_{i \in J} A_i = \bigcup_{j \in J^c} A_j \in \mathcal{F},$$

since $J^c \subset \mathbb{N}$.

(d) Let $\{B_k\}_{k \geq 1} \subset \mathcal{F}$ be given. For each $k \geq 1$, $\exists J_k \subset \mathbb{N}$ such that $B_k = \bigcup_{i \in J_k} A_i$. Let $J \equiv \bigcup_{k \geq 1} J_k \subset \mathbb{N}$. Then

$$\bigcup_{k \geq 1} B_k = \bigcup_{k \geq 1} \bigcup_{i \in J_k} A_i = \bigcup_{i \in \bigcup_{k \geq 1} J_k} A_i = \bigcup_{i \in J} A_i \in \mathcal{F},$$

since $J \subset \mathbb{N}$.

By (a), (b), (d), and the Note established in Q5 (1.4), $\mathcal{F}$ satisfies Definition 1.1.2. Hence $\mathcal{F}$ is a σ -algebra. //

Q7: Problem A

Let Ω be an uncountable set (e.g. $\mathbb{R}$) and $\mathbb{C}$ be the collection of all the singleton sets (i.e. $\mathbb{C} = \{\{\omega\} : \omega \in \Omega\}$). Find the smallest σ -algebra generated by $\mathbb{C}$. Justify your answer.

Answer

Sketch: Claim $\sigma\langle\mathbb{C}\rangle = \mathcal{F}_0 \equiv \{A \subset \Omega : A \text{ or } A^c \text{ is countable}\}$. Show $\mathcal{F}_0$ is a σ -algebra containing $\mathbb{C}$ (so $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$), then that every σ -algebra containing $\mathbb{C}$ must contain $\mathcal{F}_0$ (so $\mathcal{F}_0 \subset \sigma\langle\mathbb{C}\rangle$).

Proof.

Step 1: $\mathcal{F}_0$ is a σ -algebra.

(a) $\Omega^c = \emptyset$ is countable, so $\Omega \in \mathcal{F}_0$.

(b) Let $A \in \mathcal{F}_0$ be given. If A is countable, then $(A^c)^c = A$ is countable, so $A^c \in \mathcal{F}_0$. If A^c is countable, then $A^c \in \mathcal{F}_0$ directly. In either case $A^c \in \mathcal{F}_0$.

(d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_0$ be given. If every A_n is countable, then $\bigcup_{n \geq 1} A_n$ is a countable union of countable sets, hence countable, so $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$. Otherwise, $\exists n_0$ with $A_{n_0}^c$ countable; since

$$\left(\bigcup_{n \geq 1} A_n \right)^c = \bigcap_{n \geq 1} A_n^c \subset A_{n_0}^c,$$

and a subset of a countable set is countable, $\left(\bigcup_{n \geq 1} A_n \right)^c$ is countable, so $\bigcup_{n \geq 1} A_n \in \mathcal{F}_0$.

By (a), (b), (d) and the Note established in Q5 (1.4), $\mathcal{F}_0$ is a σ -algebra.

Step 2: $\mathbb{C} \subset \mathcal{F}_0$, hence $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$.

For any $\omega \in \Omega$, $\{\omega\}$ is finite, hence countable, so $\{\omega\} \in \mathcal{F}_0$. Thus $\mathbb{C} \subset \mathcal{F}_0$. Since $\sigma\langle\mathbb{C}\rangle$ is by definition the smallest σ -algebra containing $\mathbb{C}$, and $\mathcal{F}_0$ is a σ -algebra containing $\mathbb{C}$ (Steps 1–2), $\sigma\langle\mathbb{C}\rangle \subset \mathcal{F}_0$.

Step 3: $\mathcal{F}_0 \subset \sigma\langle\mathbb{C}\rangle$.

Let $\mathcal{F}$ be an arbitrary σ -algebra with $\mathbb{C} \subset \mathcal{F}$; it suffices to show $\mathcal{F}_0 \subset \mathcal{F}$, since $\sigma\langle\mathbb{C}\rangle$ is the intersection of all such $\mathcal{F}$.

Let $A \in \mathcal{F}_0$ be given.

Case A countable: If $A = \emptyset$, then $A = \Omega^c \in \mathcal{F}$, since $\Omega \in \mathcal{F}$ and $\mathcal{F}$ is closed under complementation. If $A \neq \emptyset$, enumerate $A = \{\omega_1, \omega_2, \dots\}$ (finite or countably infinite). Since $\mathbb{C} \subset \mathcal{F}$, $\{\omega_i\} \in \mathcal{F}$ for every i , and since $\mathcal{F}$ is a σ -algebra (closed under countable unions), $A = \bigcup_i \{\omega_i\} \in \mathcal{F}$.

Case A^c countable: By the previous case applied to A^c , $A^c \in \mathcal{F}$; since $\mathcal{F}$ is closed under complementation, $A = (A^c)^c \in \mathcal{F}$.

In either case $A \in \mathcal{F}$. Since $A \in \mathcal{F}_0$ was arbitrary, $\mathcal{F}_0 \subset \mathcal{F}$. As $\mathcal{F}$ was an arbitrary σ -algebra containing $\mathbb{C}$, $\mathcal{F}_0 \subset \mathcal{F}$ for every such $\mathcal{F}$, hence $\mathcal{F}_0 \subset \bigcap \mathcal{F} = \sigma\langle\mathbb{C}\rangle$.

By Steps 2–3, $\sigma\langle\mathbb{C}\rangle = \mathcal{F}_0 = \{A \subset \Omega : A \text{ or } A^c \text{ is countable}\}.$ //